

# Lab 2 SimpleTun Implementation

## Instructions to run:

1. Start both VMs using:
  - a. `VBoxManage startvm CS528_vm1 --type headless`
  - b. `VBoxManage startvm cs528_vm2 --type headless` (case sensitive cs528\_vm2)
2. SSH into the machines
  - a. `ssh -p 11259 cs528user@localhost` (other is 11260)
3. Run the VPN:
  - a. `cd` to minivpn(symmlink) or `cd /Desktop/cs528`
  - b. `gcc -o simpletun simpletun.c -lssl -lcrypt` on both machines
  - c. No changes in the default execution command.
4. Setup the tunnels:
  - a. `sudo ./config_vm#` (replace # by 1 or 2 as you need.)
5. Play around, have a good day!

Note: I have used mc20.cs.purdue.edu as VBoxManage was unusable in mc19.

Questions on page 2

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Results from page 6.

**Questions:**

**Question 2.1:** Think about why it is better to use UDP in the tunnel, instead of TCP write your answer in your lab report.

Answer:

UDP is faster than TCP.

Due to the following factors: lower overhead, faster transmission speed, connectionless nature facilitating NAT traversal, fault tolerance, support for multiplexing, and better performance over lossy networks.

**Question 2.2:** Think why it is not recommended to implement your own algorithm; and write that in your lab report.

Answer:

Since we implement only based on our experience, it might not be prone to errors. It doesn't mean that, if we cannot break the algorithm, someone can't.

**Question 2.3:** Why is it important for the server to release resources when a connection is broken?

Answer:

- a. If the server does not release resources, it might take up resources, eventually building up and making the server unusable.
- b. For memory resources, memory leaks may occur making the server prone to memory exploitation.
- c. For network resources, the connection pool may run out and the server will be unable to create/accept new connections.

**Snips and explanation follows...** ([page 6](#))

## Task 0: Setting up

### Importing two VMs:

```
VBoxManage import /homes/cs528/vms/cs528_vm.ova --vsys 0 --vmname CS528_vm1 --unit 9 --disk /scratch/alluri2/lab1/CS528_vm1/CS528_vm1.vmdk
```

```
VBoxManage import /homes/cs528/vms/cs528_vm.ova --vsys 0 --vmname cs528_vm2 --unit 9 --disk /scratch/alluri2/lab1/cs528_vm2/cs528_vm2.vmdk
```

**Note: For some reason, I couldn't create CS528\_vm2, so instead, I created cs528\_vm2**

**Also, on mc19.cs.purdue.edu, VBoxManage is stuck. So, I'm using mc20.cs.purdue.edu machine instead.**

### Register them:

```
VBoxManage registervm /homes/alluri2/VirtualBox\ VMs/CS528_vm1/CS528_vm1.vbox
```

```
VBoxManage registervm /homes/alluri2/VirtualBox\ VMs/cs528_vm2/cs528_vm2.vbox
```

### Setup the network:

I had a nat server setup with 192.168.1.0 and alluri2 as the name, so I went ahead and deleted it from the machine and added the following:

```
VBoxManage natnetwork add --netname alluri2 --network "192.168.15.0/24" --enable --dhcp on
```

After starting the natnetwork, the dhcp server showed up.

### Map the natnetwork to the vm:

```
VBoxManage modifyvm CS528_vm1 --nat-network1 alluri2
```

```
VBoxManage modifyvm cs528_vm2 --nat-network1 alluri2
```

### Started the **both vms** in headless mode:

```
VBoxManage startvm CS528_vm1 --type headless
```

```
VBoxManage startvm cs528_vm2 --type headless
```

### Got the information using:

```
VBoxManage guestproperty enumerate CS528_vm1 and for 2.
```

## Setup a port forwarding rule for ssh:

Find the ports: `netstat -tuln`

Select the open unused ports.

```
VBoxManage natnetwork modify --netname $USERID --port-forward-4  
"guestssh1:tcp:[]:11259:[192.168.15.4]:22"
```

```
VBoxManage natnetwork modify --netname $USERID --port-forward-4  
"guestssh2:tcp:[]:11260:[192.168.15.5]:22"
```

## Test the ports:

```
ssh -p 11259 cs528user@localhost
```

```
ssh -p 11260 cs528user@localhost
```

Password: cs528pass for both the machines.

## Clean up Shutdown the VM after use:

```
VBoxManage controlvm CS528_vm1 poweroff
```

```
VBoxManage controlvm cs528_vm2 poweroff
```

## SimpleTun (Use tmux for better UI):

Scp'ed simpletun.tar.bz2 from [backreference link](#), into desktop/simpleton.

sudo ./simpletun -i tun0 -s -d on both machines doesn't work

sudo ./simpletun -i tun0 -c 192.168.15.4 -d on client

On Machine 1 (CS528\_vm1) [192.168.15.4]:

```
sudo ip addr add 10.0.1.1/24 dev tun0
```

```
sudo ifconfig tun0 up
```

```
sudo route add -net 10.0.2.0 netmask 255.255.255.0 dev tun0
```

On Machine 2 (cs528\_vm2) [192.168.15.5]:

```
sudo ./simpletun -i tun0 -c 192.168.15.4 -d
```

```
sudo ip addr add 10.0.2.1/24 dev tun0
```

```
sudo ifconfig tun0 up
```

```
sudo route add -net 10.0.1.0 netmask 255.255.255.0 dev tun0
```

Testing the tunnel:

Ping 10.0.2.1 from machine 1.

Play with ssh: ssh 10.0.2.1 on VM1, then inside the ssh\_VM1(VM2): ssh 10.0.1.1 back to VM1 – works.

While doing this, I tried pinging other addresses in 10.0.1.0 subnet, the traffic is being registered on the log of simpletun on both executions, it's rejecting the ping request.

## Developing UDP

Tools: WinSCP syncs the src code into both the VMs from windows; watch -n 10 make & #Compiles every 10 seconds.

Password for everything: cs528pass

Get the remote ip (client ip) from the TCP handshake that already exists in simpletun.c

Establish a UDP socket using the addresses and bind the local port 55554, since TCP is defaulted to 55555.

For creating SSL certs, I've used the lab 2 instructions.

SSL verification: (I'm only using the TCP connection to get the client IP address in the server)

```

OpenSSH: client
TAP2NET 12: Read 116 bytes from the tap interface
Packet is end to end encrypted and hashed!TAP2NET 14: Written 168 bytes to the network
NET2TAP 11: Read 96 bytes from the network
Hash validated.
NET2TAP 12: Written 52 bytes to the tap interface
Hash validated.
NET2TAP 13: Read 96 bytes from the network
Hash validated.
NET2TAP 13: Written 52 bytes to the tap interface
TAP2NET 14: Read 52 bytes from the tap interface
Packet is end to end encrypted and hashed!TAP2NET 15: Written 96 bytes to the network
NET2TAP 13: Read 96 bytes from the network
Hash validated.
NET2TAP 14: Written 52 bytes to the tap interface
^C
Ctrl-C signal received. Exiting...
[03/04/2024 20:52] cs528user@cs528vm:~/minivpn$ sudo ./simpletun -d -i tun0 -s
Successfully connected to interface tun0
Init SSL
Init SSL success.
ctx success.
loaded server cert.
Enter PEM pass phrase:
loaded server private key.
Server private key valid.
loaded CA cert.
Set to peer verify on connection.
Setting up the accept BIO... done.
Setting up the incoming connection... done.
Waiting for SSL handshake... done.
Server: Waiting for TCP connectionsSERVER: Client connected from 192.168.15.5

Address is added into the network
tun0 interface is now up
route added
[03/04/2024 20:23] cs528user@cs528vm:~/minivpn$ ping 10.0.2.1
PING 10.0.2.1 (10.0.2.1) 56(84) bytes of data.
^C
--- 10.0.2.1 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3030ms

[03/04/2024 20:25] cs528user@cs528vm:~/minivpn$ ping 10.0.2.1
PING 10.0.2.1 (10.0.2.1) 56(84) bytes of data.
^C
--- 10.0.2.1 ping statistics ---
5 packets transmitted, 0 received, 100% packet loss, time 4023ms

[03/04/2024 20:26] cs528user@cs528vm:~/minivpn$ sudo ./config_vm1
Address is added into the network
tun0 interface is now up
route added
[03/04/2024 20:27] cs528user@cs528vm:~/minivpn$ ping 10.0.2.1
PING 10.0.2.1 (10.0.2.1) 56(84) bytes of data.
^C
--- 10.0.2.1 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2000ms

[03/04/2024 20:27] cs528user@cs528vm:~/minivpn$ sudo ./config_vm1
[sudo] password for cs528user:
Address is added into the network
tun0 interface is now up
route added
[03/04/2024 20:51] cs528user@cs528vm:~/minivpn$

Hash validated.
NET2TAP 11: Written 52 bytes to the tap interface
NET2TAP 11: Read 144 bytes from the network
Hash validated.
NET2TAP 12: Written 100 bytes to the tap interface
TAP2NET 10: Read 132 bytes from the tap interface
Packet is end to end encrypted and hashed!TAP2NET 11: Written 176 bytes to the network
NET2TAP 12: Read 96 bytes from the network
Hash validated.
NET2TAP 13: Written 52 bytes to the tap interface
Hash validated.
NET2TAP 13: Read 160 bytes from the network
Hash validated.
NET2TAP 14: Written 116 bytes to the tap interface
TAP2NET 11: Read 52 bytes from the tap interface
Packet is end to end encrypted and hashed!TAP2NET 12: Written 96 bytes to the network
TAP2NET 11: Read 52 bytes from the tap interface
Packet is end to end encrypted and hashed!TAP2NET 13: Written 96 bytes to the network
NET2TAP 14: Read 96 bytes from the network
Hash validated.
NET2TAP 15: Written 52 bytes to the tap interface
TAP2NET 13: Read 52 bytes from the tap interface
Packet is end to end encrypted and hashed!TAP2NET 14: Written 96 bytes to the network
^C
Ctrl-C signal received. Exiting...
[03/04/2024 20:52] cs528user@cs528vm:~/minivpn$ sudo ./simpletun -i tun0 -c 192.168.15.4 -d
Successfully connected to interface tun0
Attempting to connect to the server... Enter PEM pass phrase:
Initiating SSL handshake with the server... done.
CLIENT: Connected to server 192.168.15.4

^C
--- 10.0.1.1 ping statistics ---
13 packets transmitted, 0 received, 100% packet loss, time 12115ms

[03/04/2024 20:23] cs528user@cs528vm:~/minivpn$ sudo ./config_vm2
Address is added into the network
tun0 interface is now up
route added
[03/04/2024 20:27] cs528user@cs528vm:~/minivpn$ ping 10.0.1.1
PING 10.0.1.1 (10.0.1.1) 56(84) bytes of data.
^C
--- 10.0.1.1 ping statistics ---
1 packets transmitted, 0 received, 100% packet loss, time 0ms

[03/04/2024 20:51] cs528user@cs528vm:~/minivpn$ sudo ./config_vm2
[sudo] password for cs528user:
Address is added into the network
tun0 interface is now up
route added
[03/04/2024 20:51] cs528user@cs528vm:~/minivpn$ ping 10.0.1.1
PING 10.0.1.1 (10.0.1.1) 56(84) bytes of data.
64 bytes from 10.0.1.1: icmp_seq=1 ttl=64 time=1.22 ms
64 bytes from 10.0.1.1: icmp_seq=2 ttl=64 time=1.18 ms
^C
--- 10.0.1.1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1001ms
rtt min/avg/max/mdev = 1.180/1.202/1.224/0.022 ms
[03/04/2024 20:51] cs528user@cs528vm:~/minivpn$ ssh 10.0.1.1
cs528user@10.0.1.1's password:
[03/04/2024 20:52] cs528user@cs528vm:~/minivpn$
  
```

Finished phew!!! (UDP took 3 days. Didn't take a screenshot of just the UDP)

- I've implemented AES CBC encryption for end-to-end encryption.
- HMAC for validation.
- The key and iv are generated based on the SSL session.

I've tested the AES encryption by disabling the encryption, the communication fails. Similar for the HMAC as well.

Final result, ssh works:

```

OpenSSH SSH client
Hash validated.
NET2TAP 6: Written 52 bytes to the tap interface
NET2TAP 6: Read 1072 bytes from the network
Hash validated.
NET2TAP 7: Written 1036 bytes to the tap interface
TAP2NET 7: Read 1324 bytes from the tap interface
Packet is end to end encrypted and hashed!TAP2NET 8: Written 1360 bytes to the network
NET2TAP 7: Read 96 bytes from the network
Hash validated.
NET2TAP 8: Written 52 bytes to the tap interface
TAP2NET 8: Read 132 bytes from the tap interface
Packet is end to end encrypted and hashed!TAP2NET 9: Written 176 bytes to the network
NET2TAP 8: Read 96 bytes from the network
Hash validated.
NET2TAP 9: Written 52 bytes to the tap interface
TAP2NET 9: Read 400 bytes from the network
Hash validated.
NET2TAP 10: Written 364 bytes to the tap interface
TAP2NET 10: Read 60 bytes from the tap interface
Packet is end to end encrypted and hashed!TAP2NET 10: Written 132 bytes to the network
NET2TAP 10: Read 96 bytes from the network
Hash validated.
NET2TAP 11: Written 52 bytes to the tap interface
TAP2NET 11: Read 100 bytes from the tap interface
Packet is end to end encrypted and hashed!TAP2NET 11: Written 144 bytes to the network
NET2TAP 11: Read 96 bytes from the network
Hash validated.
NET2TAP 12: Written 52 bytes to the tap interface
NET2TAP 12: Read 144 bytes from the network
Hash validated.
NET2TAP 13: Written 100 bytes to the tap interface
TAP2NET 13: Read 132 bytes from the tap interface
Packet is end to end encrypted and hashed!TAP2NET 12: Written 176 bytes to the network
NET2TAP 13: Read 96 bytes from the network
Hash validated.
NET2TAP 14: Written 52 bytes to the tap interface
NET2TAP 14: Read 160 bytes from the network
Hash validated.
NET2TAP 15: Written 116 bytes to the tap interface
TAP2NET 15: Read 52 bytes from the tap interface
Packet is end to end encrypted and hashed!TAP2NET 15: Written 96 bytes to the network

simletun.c: In function 'main':
simletun.c:358:8: warning: assignment discards 'const' qualifier from pointer target type [enabled by default]
03/03/2024 20:19] cs528user@cs528vm:~/minivpn$ ping 10.0.
ping: unknown host 10.0.
03/03/2024 20:27] cs528user@cs528vm:~/minivpn$ sudo ./config_vm1
[sudo] password for cs528user:
tun0 interface is now up
route added
03/03/2024 20:27] cs528user@cs528vm:~/minivpn$ ping 10.0.2.1
PING 10.0.2.1 (10.0.2.1) 56(84) bytes of data:
64 bytes from 10.0.2.1: icmp_seq=1 ttl=64 time=1.45 ms
64 bytes from 10.0.2.1: icmp_seq=2 ttl=64 time=1.27 ms
64 bytes from 10.0.2.1: icmp_seq=3 ttl=64 time=1.40 ms
^C
--- 10.0.2.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 201ms
rtt min/avg/max/mdev = 1.274/1.377/1.457/0.087 ms
03/03/2024 20:27] cs528user@cs528vm:~/minivpn$ ssh cs528user@10.0.2.1
cs528user@10.0.2.1: password:
cs528user@cs528vm:~/Desktop/cs528$ gcc -o simletun simletun.c -ls1 -lcryptoclear
simletun.c: In function 'main':
simletun.c:358:8: warning: assignment discards 'const' qualifier from pointer target type [enabled by default]
/usr/bin/ld: cannot find -lcryptoclear
collect2: ld returned 1 exit status
03/03/2024 20:19] cs528user@cs528vm:~/Desktop/cs528$ gcc -o simletun simletun.c -ls1 -lcrypto
simletun.c: In function 'main':
simletun.c:358:8: warning: assignment discards 'const' qualifier from pointer target type [enabled by default]
03/03/2024 20:20] cs528user@cs528vm:~/Desktop/cs528$ sudo ./config_vm2
[sudo] password for cs528user:
Address is added into the network
tun0 interface is now up
route added
03/03/2024 20:27] cs528user@cs528vm:~/Desktop/cs528$
--- 10.0.1.1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1012ms
rtt min/avg/max/mdev = 0.947/1.044/1.141/0.097 ms
03/03/2024 16:05] cs528user@cs528vm:~/minivpn$

```

## Closing and cleaning:

```

OpenSSH session
NET2TAP 7: Read 96 bytes from the network
Hash validated.
NET2TAP 8: Written 52 bytes to the tap interface
TAP2NET 8: Read 132 bytes from the tap interface
Packet is end to end encrypted and hashed!TAP2NET 9: Written 176 bytes to the network
NET2TAP 9: Read 96 bytes from the network
Hash validated.
NET2TAP 9: Written 52 bytes to the tap interface
TAP2NET 9: Read 488 bytes from the network
Hash validated.
NET2TAP 10: Written 364 bytes to the tap interface
TAP2NET 10: Read 68 bytes from the tap interface
Packet is end to end encrypted and hashed!TAP2NET 10: Written 112 bytes to the network
NET2TAP 10: Read 96 bytes from the network
Hash validated.
NET2TAP 11: Written 52 bytes to the tap interface
TAP2NET 10: Read 160 bytes from the tap interface
Packet is end to end encrypted and hashed!TAP2NET 11: Written 144 bytes to the network
NET2TAP 11: Read 96 bytes from the network
Hash validated.
NET2TAP 12: Written 52 bytes to the tap interface
NET2TAP 12: Read 144 bytes from the network
Hash validated.
NET2TAP 13: Written 100 bytes to the tap interface
TAP2NET 11: Read 132 bytes from the tap interface
Packet is end to end encrypted and hashed!TAP2NET 12: Written 176 bytes to the network
NET2TAP 13: Read 96 bytes from the network
Hash validated.
NET2TAP 14: Written 52 bytes to the tap interface
NET2TAP 14: Read 160 bytes from the network
Hash validated.
NET2TAP 15: Written 116 bytes to the tap interface
TAP2NET 12: Read 52 bytes from the tap interface
Packet is end to end encrypted and hashed!TAP2NET 13: Written 96 bytes to the network
NET2TAP 15: Read 96 bytes from the network
Hash validated.
NET2TAP 16: Written 52 bytes to the tap interface
TAP2NET 13: Read 52 bytes from the tap interface
Packet is end to end encrypted and hashed!TAP2NET 14: Written 96 bytes to the network
Ctrl+C signal received. Exiting...
[03/03/2024 20:32] cs528user@cs528vm:~/minivpn$

simpletun.c: In function 'main':
simpletun.c:358:8: warning: assignment discards 'const' qualifier from pointer target type [enabled by default]
(03/03/2024 20:10) cs528user@cs528vm:~/minivpn$ ping 10.0.
ping: unknown host 10.0.
(03/03/2024 20:17) cs528user@cs528vm:~/minivpn$ sudo ./config_vml
[sudo] password for cs528user:
tun0 interface is now up
route added
(03/03/2024 20:27) cs528user@cs528vm:~/minivpn$ ping 10.0.2.1
PING 10.0.2.1 (10.0.2.1) 56(84) bytes of data:
64 bytes from 10.0.2.1: icmp_req=1 ttl=64 time=1.45 ms
64 bytes from 10.0.2.1: icmp_req=2 ttl=64 time=1.27 ms
64 bytes from 10.0.2.1: icmp_req=3 ttl=64 time=1.40 ms
^C
--- 10.0.2.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 201ms
rtt min/avg/max/mdev = 1.274/1.377/1.457/0.007 ms
(03/03/2024 20:27) cs528user@cs528vm:~/minivpn$ ssh cs528user@10.0.2.1
cs528user@10.0.2.1's password:
[03/03/2024 20:19] cs528user@cs528vm:~/Desktop/cs528$ gcc -o simpletun simpletun.c -lssl -lcryptoclear
simpletun.c: In function 'main':
simpletun.c:358:8: warning: assignment discards 'const' qualifier from pointer target type [enabled by default]
/usr/bin/ld: cannot find -lcryptoclear
collect2: ld returned 1 exit status
(03/03/2024 20:19) cs528user@cs528vm:~/Desktop/cs528$ gcc -o simpletun simpletun.c -lssl -lcrypto
simpletun.c: In function 'main':
simpletun.c:358:8: warning: assignment discards 'const' qualifier from pointer target type [enabled by default]
(03/03/2024 20:20) cs528user@cs528vm:~/Desktop/cs528$ sudo ./config_vml
[sudo] password for cs528user:
tun0 interface is now up
route added
(03/03/2024 20:27) cs528user@cs528vm:~/Desktop/cs528$

--- 10.0.1.1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1012ms
rtt min/avg/max/mdev = 0.947/1.044/1.141/0.097 ms
(03/03/2024 16:05) cs528user@cs528vm:~/minivpn$

```



## OpenSSL certs setup:

```
Organization Name (eg, company) [Internet Widgits Pty Ltd]:PU
Organizational Unit Name (eg, section) []:PU
Common Name (e.g. server FQDN or YOUR name) []:svr_csr
Email Address []:svr_csr@mc20.cs.purdue.edu

Please enter the following 'extra' attributes
to be sent with your certificate request
A challenge password []:cs528pass
An optional company name []:PU
mc20 113 $ openssl ca -in server.csr -out server.crt -cert ca.crt -keyfile ca.key -config openssl.cnf
Using configuration from openssl.cnf
Enter pass phrase for ca.key:
ca: ./demoCA/newcerts is not a directory
./demoCA/newcerts: No such file or directory
mc20 114 $ cd ..
mc20 115 $ openssl ca -in ./demoCA/server.csr -out ./demoCA/server.crt -cert ./demoCA/ca.crt -keyfile ./demoCA/ca.key
-config ./demoCA/openssl.cnf
Using configuration from ./demoCA/openssl.cnf
Enter pass phrase for ./demoCA/ca.key:
Check that the request matches the signature
Signature ok
Certificate Details:
  Serial Number: 4096 (0x1000)
  Validity
    Not Before: Mar  3 21:28:13 2024 GMT
    Not After : Mar  3 21:28:13 2025 GMT
  Subject:
    countryName           = US
    stateOrProvinceName   = Indiana
    organizationName       = PU
    organizationalUnitName = PU
    commonName             = svr_csr
    emailAddress           = svr_csr@mc20.cs.purdue.edu
  X509v3 extensions:
    X509v3 Basic Constraints:
      CA:FALSE
    X509v3 Subject Key Identifier:
      FE:CB:30:21:B7:0D:FF:51:4C:59:4B:9B:0D:88:A0:EC:73:BA:22:C7
    X509v3 Authority Key Identifier:
      1F:5C:F1:CE:20:2C:90:44:A8:09:10:BD:2D:82:D8:12:EE:4A:47:90
Certificate is to be certified until Mar  3 21:28:13 2025 GMT (365 days)
Sign the certificate? [y/n]:y

1 out of 1 certificate requests certified, commit? [y/n]y
Write out database with 1 new entries
Data Base Updated
mc20 116 $
```